

UTSA PROBLEM-SOLVING CLUB: OUR FAVORITE PROBLEMS OF ALL TIME

EDUARDO DUEÑEZ

The Cows of Everland 1.

In Everland's pastures, grass grows at a constant rate throughout the year. Three cows eat in two weeks all the grass on two acres of land. Two cows eat in four weeks all the grass on two acres of land. How many cows, then, will eat in six weeks all the grass on six acres of land?¹

The fork in the road 2.

A logician vacationing in the South Seas finds himself on an island inhabited by two proverbial tribes of liars and truth-tellers. Members of one tribe always tell the truth, members of the other always lie. He comes to a fork in a road and has to ask a native bystander which branch he should take to reach a village. He has no way of telling whether the native is a truth-teller or a liar. The logician thinks a moment, then asks one question only. From the reply he knows which road to take. What question does he ask?

The Mutilated Chessboard 3.

If you have a regular 8×8 chessboard, it can be tiled (covered) by 32 dominoes 2-by-1 in size; it can even be done in various ways, but certainly by laying down 4 dominoes in each of the 8 rows of the board. However, your puppies Rowdy and Runnie took up the task of mauling the poor chessboard, tugging at it from two opposite corners, each biting off one corner square (on opposite ends of a chessboard diagonal). Well, so much for tiling... Or, perhaps can we still tile the mutilated chessboard using 31 dominoes? If so, how?

Sandwiches! 4.

You eat sandwiches. You love sandwiches. No: you *love* sandwiches.² But *only* mini-sandwiches that are exactly $2\text{ in} \times 2\text{ in} \times 1\text{ in}$ in size.³ When visiting Mexico last summer, you realized that, delicious as the tacos were south of the border, you *missed* your sandwiches, so naturally you asked mom to ship you some. She packed full and sent you a cubic box $20\text{ in} \times 20\text{ in} \times 20\text{ in}$ in size (a total of 2000 mini-sandwiches!). Given the urgency and sheer volume of the task, when it came to packing, all sandwiches ended up in random positions: some facing up or down down, some facing either of the four other sides of the box. When going through customs, an evil inspector took a very long needle and pierced right through the box (ouch!). Is there any chance that the needle may have missed every single sandwich (somehow finding its way right through their edges), or is it the case that necessarily *some* sandwich(es) must have been pierced by the needle?

¹The classic version of this problem (involving larger numbers and possibly goats instead of cows, but an otherwise identical setup) is (apocryphally) attributed to Newton.

²Between 2018 and 2014, sandwiches—not pizza—were the snack of choice at the UTSA PSC.

³Okay, truthfully, we like sandwiches of all shapes and sizes, but for the sake of this problem let's pretend we are very picky.